CareRoute-X — AI/ML Architecture & Data Specification

Comprehensive Technical Reference: Datasets, Supervised Models, LLM Triage, and Real-Life Clinical Case Flows

Executive Architecture Summary: CareRoute-X employs a hybrid, multi-layered intelligence engine designed specifically for hospital outpatient care navigation. It integrates:
1) **Supervised Machine Learning:** A calibrated TF-IDF + Logistic Regression model trained on Symptom2Disease for specialty classification.
2) **Generative AI (LLMs):** Google Gemini structured inference for conversational symptom intake, negation extraction, and empathetic inquiry.
3) **Medical Ontologies:** The DDXPlus evidence ontology dataset for deterministic symptom validation and dependency management.
4) **Operational Statistical Models:** Bayesian shrinkage for doctor rating reliability and deterministic wait-time formulations.

1. Classical Supervised Machine Learning: Symptom-to-Specialty Classifier

The primary supervised learning component operates in backend/ml/. It classifies raw patient-described symptoms into clinical disease categories and maps them to appropriate hospital departments.

1.1 Dataset: Symptom2Disease (ml/data/Symptom2Disease.csv)

- **Volume & Balance:** 1,200 clinical symptom descriptions across 24 balanced disease categories (50 samples per class).
- **Classes Covered:** Acne, Arthritis, Bronchial Asthma, Cervical Spondylosis, Chicken Pox, Common Cold, Dengue, Hemorrhoids, Fungal Infection, Hypertension, Impetigo, Jaundice, Malaria, Migraine, Pneumonia, Psoriasis, Typhoid, Varicose Veins, Allergy, Diabetes, Drug Reaction, GERD, Peptic Ulcer Disease, and UTI.
- **Audit & Integrity:** Stratified 70% Train (840) / 15% Validation (180) / 15% Test (180) split with verified zero empty/null rows and verified cross-split leakage checks.

1.2 Model Architecture & Mathematical Calibration

1. Feature Extraction: TfidfVectorizer(ngram_range=(1, 2), max_features=5000, sublinear_tf=True, stop_words='english'). Fitted strictly on X_train to prevent data leakage.
2. Base Classifier: Multinomial LogisticRegression(C=3.0, max_iter=1000, class_weight='balanced').
3. Probability Calibration (Platt Scaling): Standard logistic regression outputs are often overconfident. Platt scaling trains a sigmoid regressor over 5-fold cross-validation (CalibratedClassifierCV(method='sigmoid', cv=5)), mapping raw logits into true empirical probabilities.

Performance Metric	Uncalibrated Baseline	Calibrated Model (Production)	Clinical Significance
Top-1 Accuracy	94.44%	**95.00%**	Accurately identifies primary suspected disease.
Top-3 Accuracy	N/A	**99.44%**	Ensures correct condition is in differential list.
Brier Score Loss	0.2894	**0.1475 (49% drop)**	Lower score indicates superior probability accuracy.
Expected Calibration Error (ECE)	0.4004	**0.2326 (42% drop)**	Confidence directly mirrors empirical truth.
Abstention Threshold	None	**p < 0.25 Defers**	Low-confidence cases safely defer to human triage.

1.3 Clinical Specialty Mapping Matrix (DISEASE_TO_SPECIALTY)

Specialty Code	Department Name	Mapped Diagnostic Classes
ORTHO	Orthopedics & Spine	Arthritis, Cervical Spondylosis
CARDIO	Cardiology & Vascular	Hypertension, Varicose Veins
RESPIRATORY	Pulmonology	Bronchial Asthma, Pneumonia
DERM	Dermatology	Acne, Chicken Pox, Fungal Infection, Impetigo, Psoriasis, Drug Reaction
NEURO	Neurology	Migraine
GENERAL	Internal Medicine	Common Cold, Dengue, Jaundice, Malaria, Typhoid, Diabetes, GERD, Peptic Ulcers, UTI

2. Generative AI Engine: Conversational Clinical Triage (Google Gemini)

Implemented in backend/app/services/llm/gemini.py and conforming to the LLMProvider protocol, Google Gemini handles dynamic, conversational patient intake.

2.1 Core LLM Operational Functions

- **extract_patient_evidence(message, context):** Uses structured JSON output schemas to parse raw patient narratives into typed clinical entities: Chief Complaint, Duration, Severity Score (1-10), Numerical Vitals (e.g., Temp 102°F), Extracted Symptoms, and Pertinent Negatives (symptoms the patient explicitly denies).
- **propose_next_question(context):** Analyzes current gathered evidence against clinical uncertainty. Returns an AdaptiveQuestionProposal containing next question text, clinical intent, target symptom, and whether human review is mandated.
- **phrase_question(controlled_question, evidence_id):** Transforms dry clinical inquiries (e.g., *'Presence of dysuria'*) into patient-friendly, empathetic conversational phrasing (e.g., *'Do you feel any burning sensation when urinating?'*).
- **summarize_conversation(messages):** Synthesizes multi-turn dialogue into a concise physician handoff note prior to the outpatient consultation.

2.2 Resilience, Cost & Safety Hardening (Phase 3B.6)

• **Deterministic Extraction Bypass:** Exact affirmative or negative replies to binary questions (e.g., 'Yes', 'No') are validated directly against ADAPTIVE_EVIDENCE_REGISTRY, completely bypassing LLM inference and saving 50% of API requests.
• **Pre-Question Stopping Evaluation:** TriageStoppingPolicy.should_stop_pre_question() checks terminal intake conditions (emergency red flags, patient opt-out, or max 7 turns reached) *before* proposing the next question, saving 1 expensive LLM call per terminal session.
• **Calibrated Token Windows:** Response token caps were tuned down from 1500 to 700 tokens for evidence extraction and 500 tokens for question proposal.
• **Strict Quota Fast-Fail:** Permanent HTTP 429 quota errors fail fast (< 5ms) without futile sleep loops, immediately offering the patient manual doctor selection.

3. Clinical Knowledge Base & Ontology: DDXPlus Evidence Dataset

3.1 The Dataset: backend/data/release_evidences.json

CareRoute-X integrates the **DDXPlus clinical benchmark** evidence dataset. It contains hundreds of validated medical evidence codes, data types (Binary, Categorical, Numerical), and hierarchical parent-child relationships.

3.2 The Ontology Engine: DDXPlusEvidenceProvider

• **In-Memory Lookup:** Singleton instance that loads and indexes symptoms, ensuring zero-latency access during active conversational turns.
• **Parent-Child Dependency Enforcement:** Sub-questions (e.g., *'Character of chest pain'*) are only activated if their parent symptom (*'Chest pain'*) is validated as PRESENT.
• **Value Domain Verification:** Rejects hallucinated or invalid LLM extractions that do not conform to canonical medical standards.

4. Statistical & Operational Optimization Models

4.1 Bayesian Shrinkage for Doctor Rating Reliability

To prevent a new doctor with a single 5-star review outranking an experienced physician with 150 reviews averaging 4.8 stars, CareRoute-X applies Bayesian shrinkage toward a hospital prior:
Formula: S_rating = (v * R + m * C) / (v + m) (where v = review count, R = average rating, C = baseline 4.0, m = smoothing strength 10).

4.2 Dynamic Operational Queue Wait-Time Formulation

Derived from live consultation state transitions in QueueService:
Formula: Estimated_Wait_Minutes = max(5, patients_ahead * 15)
Dynamically updates as preceding patients transition through WAITING → CALLED → IN_CONSULTATION → COMPLETED.

5. End-to-End Real-Life Clinical Example Walkthrough

The following real-life scenario illustrates how all AI/ML models, datasets, and operational algorithms interact seamlessly during an outpatient visit.

Stage 1: Unstructured Patient Input

Patient (Sarah, 42): *"I've had severe throbbing pain on the right side of my head since yesterday morning. Bright light and loud noises make it much worse, and I feel nauseous. Painkillers didn't help."*

Stage 2: Gemini LLM Structured Extraction (gemini.py)

• **Chief Complaint:** Unilateral throbbing headache.
• **Duration:** ~24-36 hours.
• **Severity:** 8 / 10.
• **Extracted Evidences:** photophobia=PRESENT, phonophobia=PRESENT, nausea=PRESENT.
• **Emergency Red-Flag Check:** No sudden onset 'thunderclap' headache, no fever, no neck stiffness, no neurological focal deficits → **Red Flag: FALSE**.

Stage 3: ML Classifier & Specialty Routing (train.py)

• **TF-IDF Vectorization:** Matches n-grams: *'throbbing pain'*, *'head'*, *'bright light'*, *'nauseous'*.
• **Calibrated Model Prediction:** Top-1 Condition = **Migraine (p = 0.942)**; Top-2 = Tension Headache (p = 0.041).
• **Specialty Lookup:** DISEASE_TO_SPECIALTY['Migraine'] → **NEURO (Neurology & Headache Clinic)**.

Stage 4: Adaptive Clinical Inquiry (orchestrator.py)

Orchestrator invokes propose_next_question(). Gemini proposes: *'Are you experiencing any visual disturbances such as flashing lights or blind spots (aura)?'*
Patient replies: *'No'* → Bypasses LLM extraction via ADAPTIVE_EVIDENCE_REGISTRY, recording visual_aura=ABSENT.

Stage 5: Explainable Doctor Ranking (backend/app/ranking/)

Candidates filtered by Hospital boundary + NEURO specialty:
• **Dr. Alisha Rao, MD (Neurologist):**
- Specialty Match: 1.0 (weight 0.30) → 0.30
- Slot Availability: 2 open slots today (weight 0.20) → 0.20
- Queue Wait: 1 ahead (15 mins) (weight 0.15) → 0.13
- Experience: 16 yrs / 30 (weight 0.10) → 0.053
- Bayesian Rating: 4.8 stars from 62 reviews (shrinkage: 4.69) (weight 0.10) → 0.094
- **Composite Score: 0.877 (Rank #1)**

Stage 6: Clinical Handoff & Live Queue

Sarah books Dr. Rao. Queue token **#N-04** issued.
• Sarah monitors wait time (15 mins) remotely in the hospital lobby.
• Dr. Rao's portal displays Gemini's pre-consultation summary: *'42yo female presenting with unilateral throbbing headache (8/10), photophobia, nausea. Pertinent negatives: No aura, no fever, no thunderclap onset. Suspected acute migraine.'*